



Introduction to AJAX

1. Need of AJAX
2. Advantages of AJAX
3. Disadvantages of AJAX
4. Using AJAX in Javascript

1. Need of AJAX

- AJAX stands for Asynchronous JavaScript and XML.
- Ajax helps to transport data to client asynchronously without refreshing the web page.

Synchronous requests	Asynchronous requests
Client need to wait for the response	Client need not wait for the response
Entire page gets refreshed	Only part of the page gets refreshed

- Examples:
 - Typing a search query in google, pops ups few suggestions
 - Populating a drop down for selecting cities based on selected state
 - Getting live scores in the web page

- AJAX Technologies include
 - **JavaScript:** When an event occurs in a website, a JavaScript function is executed.
 - **HTML/XHTML and CSS:** Responsible for the presentation of the web content
 - **DOM:** DOM is an API that is used for accessing the existing content and updating content after receiving response.
 - **XMLHttpRequest:** It is used to establish asynchronous communication with the server.
 - **XML or JSON:** JSON is lightweight and faster than XML. It is used as format of response

- **Improves response time:** web page's visitors will never have to see a white window and wait for pages to refresh
- **Enabled asynchronous calls:** Allows to make a request in the background
- **Enhanced User Experience:** responsive web pages gives richer experience to the end user
- **Reduced server load:** Only required amount of data gets transferred
- **Form validation:** validity of data in the form can be checked before the form is submitted

- If any customer has disabled JavaScript, AJAX requests will not work
- Makes web page more complex to debug
- Ajax requests can not be registered with the browser's history
- AJAX requests can not be bookmarked
- Search engines can not index AJAX content
- Web pages with AJAX are vulnerable to web attacks

Basic steps:

- Some event occurs which triggers all further steps to be carried for sending asynchronous request to the server.
- Creating object which represents AJAX object holding information about asynchronous request and response cycle
- Initialize the request by specifying request URL and method
- Send the request from client side
- Client should check whether response is received from server side by using onreadystatechange property
- Update the web page using HTML DOM functions when entire response is received from server side.